Econometrics 120A: Discussion Section

Bivariate Distributions and Sampling

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Joint, marginal and conditional distributions

Recap: joint and marginal distributions

- **Joint distribution:** $P[X = x,\ Y = y]$ for every pair (x, y).
- **Valid** if ❶ every entry ≥ 0 and ❷ all entries sum to 1.
- **Marginals:** sum out the other variable

$$P[X = x] = \sum_{\text{all } y} P[X = x, Y = y]$$

$$P[Y = y] = \sum_{\text{all } x} P[X = x, Y = y]$$

- Joint $\Rightarrow$ marginals: always possible. Marginals $\Rightarrow$ joint: only under independence.

$P(S=0,Y=0)=\frac{1}{4}$, $P(S=0,Y=1)=0$
$P(S=0)=\sum_y P(S=0,Y=y)=\frac{1}{4}$

$S \backslash Y$	0	1	$P(s)$
0	1/4	0	**1/4**
1	1/4	1/4	**1/2**
2	0	1/4	**1/4**
$P(y)$	**1/2**	**1/2**	1

Lecture Example 2: two coin tosses, S = number of heads, Y = second toss.

Row sums $\rightarrow$ marginal of S.
Column sums $\downarrow$ marginal of Y.

Recap: conditional distributions and independence

conditional prob. $= \frac{\text{joint prob.}}{\text{marginal prob.}}$

- **Conditional:** take one slice, rescale so it sums to 1

$$P[X = x \mid Y = y] = \frac{P[X = x,\ Y = y]}{P[Y = y]}$$

- **Conditional mean:**
  $E[X \mid Y = y] = \sum_x x\, P[X = x \mid Y = y]$
- **Independence:** for *all* (x, y),
  $P[X = x, Y = y] = P[X = x]\, P[Y = y]$,
  equivalently
  $P[X = x \mid Y = y] = P[X = x]$.
- One failing cell breaks independence.
- Independent $\neq$ mutually exclusive.

$S \backslash Y$	0	1	$P(s)$
0	1/4	0	1/4
1	1/4	1/4	1/2
2	0	1/4	1/4
$P(y)$	1/2	1/2	1

Slice $S = 1$:

$$P[Y = 0 \mid S = 1] = \frac{1/4}{1/2} = \frac{1}{2}$$

Independence check:
$P[S = 2, Y = 0] = 0$, but
$P[S = 2]\, P[Y = 0] = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$
$\Rightarrow$ S and Y are **not** independent.

Conditional Distribution Questions

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(a) What is the conditional distribution of X given that $Y = 1$?

(b) What is the mean of X given $Y = 1$?

(c) What is the variance of X given $Y = 1$?

(d) Suppose we are trying to learn the mean of X, but observe X only when $Y = 1$ (e.g. X = number of holidays a person takes per year, Y = whether their phone number is listed, so they can be included in the survey). Otherwise the survey is well run. What is the bias in our estimate of the mean of X?

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(a) What is the conditional distribution of X given that $Y = 1$?

Marginal distn $P(Y=1) = \sum_x P(X=x, Y=1)$

$= P(X=0, Y=1) + P(X=1, Y=1) + P(X=2, Y=1)$

$= 0 + 0.4 + 0.1$

$= 0.5$

Conditional distn

$X \mid Y=1$	$P(X=x \mid Y=1)$
0	0
1	0.8
2	0.2

$P(X=0 \mid Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{0}{0.5} = 0$

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(b) What is the mean of X given $Y = 1$?

Conditional dist'n

$X \mid Y=1$	$P(X=x \mid Y=1)$
0	0
1	0.8
2	0.2

$$\mathbb{E}[X \mid Y=1] = \sum_x x \cdot P(X=x \mid Y=1) = 0\times 0 + 1\times 0.8 + 2\times 0.2 = 1.2$$

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(c) What is the variance of X given $Y = 1$?

Conditional dist'n

$X \mid Y=1$	$P(X=x \mid Y=1)$
0	0
1	0.8
2	0.2

Method 1: $\mathrm{Var}(X \mid Y=1) = \sum_x \left(x - E[X \mid Y=1]\right)^2 P(X=x \mid Y=1)$

$= (0-1.2)^2 \cdot 0 + (1-1.2)^2 \cdot 0.8 + (2-1.2)^2 \cdot 0.2$

$= 0.04 \times 0.8 + 0.64 \times 0.2$

$= 0.16$

Method 2:

$X^2 \mid Y=1$	$P(X^2 \mid Y=1)$
0	0
1	0.8
4	0.2

$E[X^2 \mid Y=1] = 1.6$

$\mathrm{Var}(X^2 \mid Y=1) = E[X^2 \mid Y=1] - \left(E[X \mid Y=1]\right)^2$

$= 1.6 - 1.2^2$

$= 1.6 - 1.44$

$= 0.16$

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1	$p(X=x)$
0	0.1	0	0.1
1	0.4	0.4	0.8
2	0	0.1	0.1

Satisfaction score of a product

Wether the customer file a claim or not

(d) We observe X only when $Y = 1$ (only people with a listed number get surveyed). What is the bias in our estimate of the mean of X?

$\mathbb{E}[X|Y=1]=1.2$ from part (b)

$\mathbb{E}[X]=1.0$.

We obtain the estimate for conditional mean rather than the unconditional mean. (The bias)

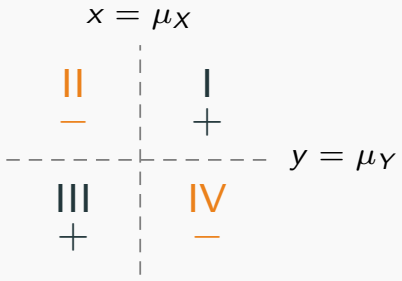
Covariance and correlation

Recap: from sample to population

- DS4, sample: $s_{xy} = \frac{1}{n-1}\sum_i (x_i - \bar{x})(y_i - \bar{y})$
- Population: weight each pair by its probability

$$\mathrm{Cov}(X, Y) = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) \times P[X = x, Y = y]$$

- Shortcut: $\mathrm{Cov}(X, Y) = E[XY] - E[X]\,E[Y]$
- Correlation: $\rho_{XY} = \dfrac{\mathrm{Cov}(X, Y)}{\sigma_X\,\sigma_Y}$, unitless, $-1 \le \rho \le 1$
- Independent $\Rightarrow$ Cov $= 0$.
  Cov $= 0$ $\not\Rightarrow$ independent: covariance only picks up *linear* co-movement.



Same quadrant logic as DS4, but each (x, y) now counts with weight $P[X = x, Y = y]$.

Lecture Example 2:
$\mathrm{Cov}(S, Y) = \frac{1}{4}$,

$$\rho = \frac{1/4}{\sqrt{1/2}\,\sqrt{1/4}} \approx 0.71$$

Let $Y = X^2$, and X w/ the following dist'n

X	Y	P(X=x)
−1	1	1/3
0	0	1/3
1	1	1/3

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y] = -\frac{2}{9} + \frac{2}{9} - 0 \cdot \frac{2}{3} = 0.$

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(a) Compute the mean of X and the mean of Y.

(b) Compute the covariance of X and Y.

(c) Without doing additional math, what does this result suggest regarding the relationship between X and Y? Are they related in any way?

(d) What is the mean of $g(X, Y) = Y/X$?

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(a) Compute the mean of X and the mean of Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

$P(X=1)=0.3$
$P(X=2)=0.4$
$P(X=3)=0.3$

$3+8+9$

$E[X]=1\times0.3+2\times0.4+3\times0.3=2$

$P(Y=0)=0.4$ $P(Y=1)=0.6$ $E[Y]=0.6$

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(b) Compute the covariance of X and Y.

$$\mathrm{Cov}(X,Y) = \sum_{x,y} P(X=x, Y=y)\,(x - \mathbb{E}[X])(y - \mathbb{E}[y])$$

$$= 0.1 \cdot (1-2)\cdot(0-0.6) + 0.2\cdot(1-2)(1-0.6)$$
$$+0.2\,(2-2)\cdot(0-0.6) + 0.2\,(2-2)(1-0.6)$$
$$+0.1\,(3-2)(0-0.6) + 0.2\,(3-2)(1-0.6)$$
$$= (-0.1)\times(-0.4) + (-0.2)\times 0.4 + 0.1\times(-0.4) + 0.2\times(0.4)$$
$$= 0$$

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(c) Without doing additional math, what does this result suggest about the relationship between X and Y? Are they related in any way?

$Cov(X,Y)=0$ doesn't necessarily imply $X \perp\!\!\!\perp Y$ (X is independent of Y).

We can claim that on average higher values for X are not related to higher value of Y.

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(d) What is the mean of $g(X, Y) = Y/X$?

Let's define a random variable $R = \frac{Y}{X}$

$Y \in \{0, 1\}$
$X \in \{1, 2, 3\}$

So $R = \left\{ \frac{0}{1}, \frac{0}{2}, \frac{0}{3}, \frac{1}{1}, \frac{1}{2}, \frac{1}{3} \right\} = \left\{ 0, 1, \frac{1}{2}, \frac{1}{3} \right\}$

$0.1 \quad 0.2 \quad 0.1 \quad 0.2 \quad 0.2 \quad 0.2$

R	$P(R=r)$
0	0.4
$\frac{1}{3}$	0.2
$\frac{1}{2}$	0.2
1	0.2

$$E[R] = \frac{0.2}{3} + 0.1 + 0.2 = \frac{2+3}{30} + 6 = \frac{1}{30}$$

Functions of two random variables

Recap: means and variances of $g(X, Y)$

- New random variable $R = g(X, Y)$. Two ways to get $E[R]$:
  1. Find the distribution of R, then $E[R] = \sum_r r\, P[R = r]$
  2. Directly: $E[g(X, Y)] = \sum_x \sum_y g(x, y)\, P[X = x, Y = y]$

  Same answer; pick whichever is easier.
- In general $E[g(X, Y)] \neq g(E[X], E[Y])$. Linear functions are the exception:

Mean	$E[aX + bY] = aE[X] + bE[Y]$
Variance	$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\, \mathrm{Cov}(X, Y)$

- The mean rule always holds, independent or not.
- If X, Y are independent the highlighted term is 0.
- With $a = b = \frac{1}{2}$ we get the average of two random variables: the key step for sampling.

Problem Set 3, Question 5

Suppose we have the joint distribution between two random variables X and Y such that

$X \backslash Y$	0	1
-1	$3/8$	0
0	$1/8$	$3/8$
1	0	$1/8$

(a) We want to create a new random variable $X + Y$. Compute the mean of $X + Y$.

(b) Compute the variance of $X + Y$.

Problem Set 3, Question 5

Suppose we have the joint distribution between two random variables X and Y such that

$X \backslash Y$	0	1	
-1	3/8	0	$\frac{3}{8}$
0	1/8	3/8	$\frac{1}{2}$
1	0	1/8	$\frac{1}{8}$
	$\frac{1}{2}$	$\frac{1}{2}$	

(a) Compute the mean of the new random variable $X + Y$.

Method 1 $E[X+Y] = \sum_{(x,y)} P(X=x, Y=y)(x+y)$ $\quad g(X,Y)=X+Y$

$= P(X=-1,Y=0)(-1+0) + \cdots + P(X=1,Y=1)(1+1)$

$= \frac{3}{8}(-1) + 0(0) + \frac{1}{8}(0) + \frac{3}{8}(1) + 0(1) + \frac{1}{8}(2)$

$= \frac{1}{4}$

Method 2 $X+Y \in \{-1+0, -1+1, 0+0, 0+1, 1+0, 1+1\} = \{-1, 0, 0, 1, 1, 2\}$

$X+Y$	$P(X+Y)$
-1	$\frac{3}{8}$
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{1}{8}$

$E[X+Y] = -\frac{3}{8} + 0 + \frac{3}{8} + \frac{2}{8} = \frac{1}{4}$

Method 3. $E[X+Y] = E[X] + E[Y]$

$= \left(-\frac{3}{8} + \frac{1}{8}\right) + \frac{1}{2} = \frac{1}{4}$

Problem Set 3, Question 5

Suppose we have the joint distribution between two random variables X and Y such that

$X \backslash Y$	0	1
-1	3/8	0
0	1/8	3/8
1	0	1/8

(b) Compute the variance of $X + Y$.

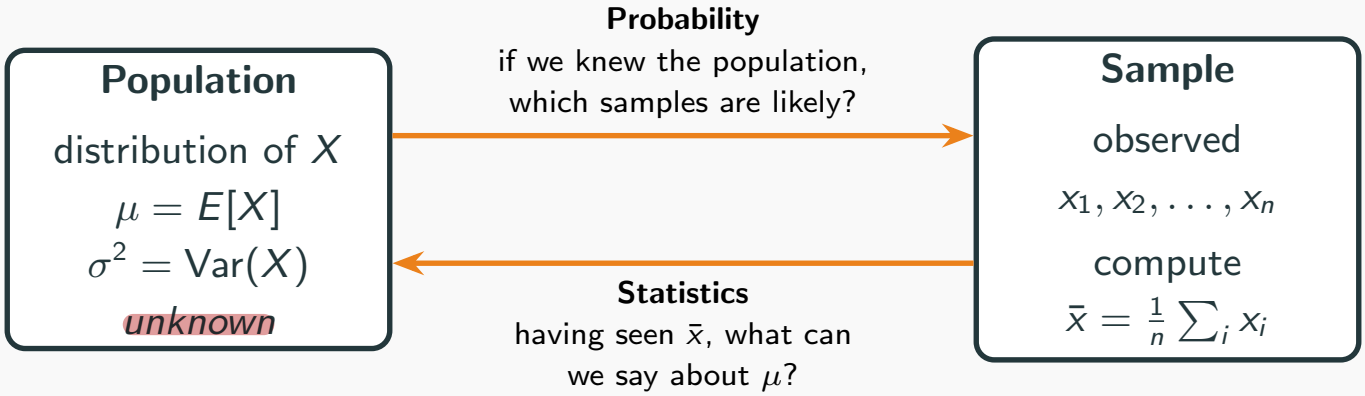
$$\mathrm{Var}(X+Y) = \mathbb{E}\left(X+Y - \underbrace{\mathbb{E}[X+Y]}_{=\frac{1}{4}}\right)^2 = \sum_{(x,y)} P(X=x, Y=y)\left(x+y-\tfrac{1}{4}\right)^2$$

$$= \frac{3}{8}\left(-1+0-\tfrac{1}{4}\right)^2 + \cdots + \frac{1}{8}\left(1+1-\tfrac{1}{4}\right)^2$$

$$= \frac{152}{128}$$

Sampling

The big picture



Sample average
random! depending on the sample drawn

- ⇨ The bridge between the two: the distribution of $\bar{X}$ (the **sampling distribution**).
- ⇨ It tells us which sample averages are likely if our model of the population is right, so we can check the model against the data.

A sample is n random variables

- **Before** we collect data, each observation is a random variable: $X_1, X_2, \ldots, X_n$.
  After, we see numbers: $x_1, x_2, \ldots, x_n$.
- Poll example: $X_i = 1$ if person i supports the candidate, 0 otherwise, so $X_i \sim \text{Bernoulli}(p)$.
- **Very Simple Random Sample (VSRS):**
  1. **Independent:** $P[X_1 = x_1, X_2 = x_2] = P[X_1 = x_1]\, P[X_2 = x_2]$
  2. **Identically distributed:** every X_i has the same distribution, so the same μ and σ^2
- So a sample is just a big joint distribution. Everything from the first half of today applies, with n variables instead of two.

Moments of the Sample Mean

Using the linear combination rules

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \text{(a linear combination with weights } 1/n\text{)}$$

Mean $$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}\, n\mu = \mu \qquad \text{(linearity only)}$$

Variance $$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\Big[\underbrace{\textstyle\sum_i \mathrm{Var}(X_i)}_{n\sigma^2 \text{ (identical)}} + \underbrace{\textstyle\sum_{i\neq j} \mathrm{Cov}(X_i, X_j)}_{0 \text{ (independent)}}\Big] = \frac{\sigma^2}{n}$$

- $n = 2$: $\mathrm{Var}\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 + 2\cdot\frac{1}{4}\cdot 0 = \frac{\sigma^2}{2}$ (the $\mathrm{Var}(aX + bY)$ rule with $a = b = \frac{1}{2}$).
- **Warning:** σ = spread of *one* observation; $\sigma/\sqrt{n}$ = spread of the *average*. Ask yourself: is the question about one person, or about an average?

The Shape of $\bar{X}$

Three results

if individual observations follow a bell-shaped normal dist'n, then the sample average is exactly normal dist'n.

1. **Normal data, exact:** if $X_i \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N(\mu, \sigma^2/n)$ for any n.
2. **Law of Large Numbers:** for any $\varepsilon > 0$, $P\left[|\bar{X}_n - \mu| > \varepsilon\right] \to 0$ as $n \to \infty$. As we collect more observations, the sample average gets closer to the truth.
   $\text{Var}(\bar{X}) = \sigma^2/n \to 0$, so $\bar{X}$ piles up on μ. This is why we average.
3. **Central Limit Theorem:** if $X_i \sim (\mu, \sigma^2)$, *any* distribution, then for large n

As we collect more observations, the sample average becomes approximately normal even if the individual observation is not normally distributed.

$$\bar{X}_n \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

$\overset{a}{\sim}$ "approximately distributed as"

Recipe (same normal table as DS4):

$$P[\bar{X} > c] \approx P\left[Z > \frac{c - \mu}{\sigma/\sqrt{n}}\right], \qquad Z \sim N(0,1)$$

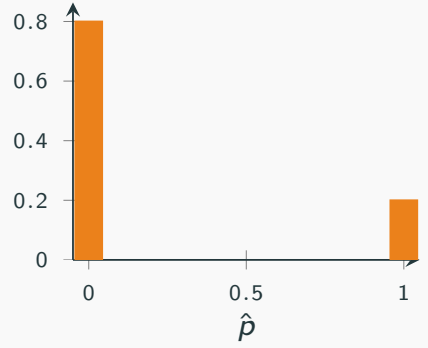
CLT for Bernoulli random variables

unobserved, need to be estimated
Suppose $p=0.2$ but we don't know

$X_i \sim \text{Bernoulli}(p)$: $\mu = p$, $\sigma^2 = p(1-p)$ $\Rightarrow$

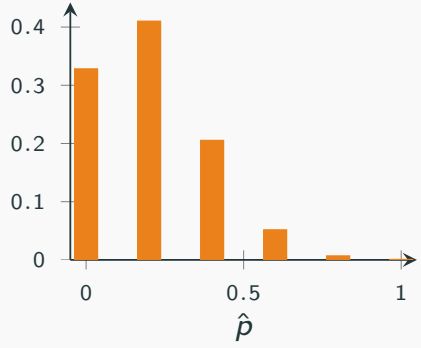
$$\hat{p} = \bar{X} \overset{a}{\sim} N\left(p, \frac{p(1-p)}{n}\right)$$

if we repeat 1 trial many times
80% produce $\hat{p}=0$, 20% produce $\hat{p}=1$



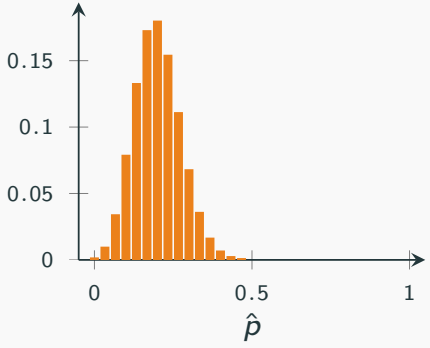
if we repeat 5 trials

$n = 5$



if we repeat 30 trials
peak at $\hat{p}=0.2$

$n = 30$



Exact distribution of $\hat{p} = S/n$ with $S \sim \text{Binomial}(n, 0.2)$. Skewed at $n = 1$; by $n = 30$ it is bell shaped and concentrated near $p = 0.2$.

Lecture example: $n = 400$, $p = 0.6$: $\sigma_{\hat{p}} = \sqrt{0.6 \cdot 0.4/400} \approx 0.0245$.

Sampling Exercise

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(a) What are $E[\bar{X}]$ and the standard deviation of $\bar{X}$?

(b) Hours are not normally distributed. Why can we still use the normal distribution for $\bar{X}$? What is its approximate distribution?

(c) Approximate the probability that the sample average is above 14 hours.

(d) A classmate concludes: "So only about 2% of students study more than 14 hours a week." What is wrong with this?

(e) Redo (c) with $n = 144$. Which result from lecture does the change illustrate?

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(a) What are $E[\bar{X}]$ and the standard deviation of $\bar{X}$?

$E[\bar{X}] = \mu = 12$

$Var(\bar{X}) = \frac{\sigma^2}{n} = \frac{36}{36} = 1 \Rightarrow$ S.D. $(\bar{X}) = 1$ hour

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

but average hour is normally distributed under CLT.

(b) Hours are not normally distributed. Why can we still use the normal distribution for $\bar{X}$? What is its approximate distribution?

Because Central Limit Theorem states that for a sufficiently large sample, the sampling distribution of the average is approximately normal.

$$\bar{X} \overset{a}{\sim} N(12, 1)$$

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(c) Approximate the probability that the sample average is above 14 hours.

z	0.00	0.01	0.02	0.03	0.04
1.9	0.0287	0.0281	0.0274	0.0268	0.0262
2.0	0.0228	0.0222	0.0217	0.0212	0.0207
2.1	0.0179	0.0174	0.0170	0.0166	0.0162

Table: $P(Z > z)$

$\bar{X} \overset{a}{\sim} N(12, 1)$

$P(\bar{X} > 14) = P\left(\frac{\bar{X}-12}{1} > \frac{14-12}{1}\right) = P(Z > 2) \approx 0.0228$

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(d) A classmate concludes: "So only about 2% of students study more than 14 hours a week." What is wrong with this?

$P(X_i > 14) = ?$ we don't know the dist'n of X

The calculation in part (c) is about the average of 36 students not an individual student.

$P(\bar{X}_n > 14)$, $n = 36$

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(e) Redo (c) with $n = 144$. Which result from lecture does the change illustrate? $\mathrm{Var}(\bar{X}_{144}) = \frac{36}{144} = \frac{3}{12} = \frac{1}{4}$

$\bar{X} \overset{a}{\sim} N(12, \frac{1}{4})$

$P(\bar{X} > 14) = P\left(\frac{\bar{X}-12}{\sqrt{\frac{1}{4}}} > \frac{14-12}{\frac{1}{2}}\right) = P(Z > 4) \approx 0.000031$

It illustrates Law of Large Numbers: larger samples produce averages more concentrated around the population mean.

Summary

- **Joint $\to$ marginal:** sum out the other variable. **Conditional:** slice and rescale.
- **Independence** must hold in *every* cell; equivalently, every conditional equals the marginal.
- **Covariance** $= 0$ does not mean independent; a constant conditional mean does not either.
- $E[aX + bY] = aE[X] + bE[Y]$ always; $\text{Var}(aX + bY)$ needs the $2ab\,\text{Cov}$ term.
- A random sample is n independent, identically distributed random variables, so $E[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \sigma^2/n$.
- **LLN:** $\bar{X}$ gets close to μ. **CLT:** $\bar{X} \overset{a}{\sim} N(\mu, \sigma^2/n)$ whatever the distribution of X_i.